Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Need for statistics: purposes, and the collection and tabulation of data. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	9
Topic	Need for statistic
Subtopic	Need for statistics: purposes, and the collection and tabulation of data
Theme	—
Curriculum objectives	• explain the purpose of statistic • analyse and interpret statistical data presented in tables
Previous knowledge	• Students are familiar with counting and recording information.
Lesson sequence	• explain the purpose of statistic — the teacher introduces the concept of statistics as a method of collecting and organising information • analyse and interpret statistical data presented in tables — the teacher takes the students through collecting data, recording it with tally marks, and setting it out in a frequency table
Instructional materials	• classroom register, bottle tops, pencils or counters.
Evaluation	• (a) Define statistic • (b) State three methods of collecting data. • (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

Lesson objectives

1.	explain the purpose of statistic
2.	analyse and interpret statistical data presented in tables

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Need for statistics: purposes, and the collection and tabulation of data

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: What is Statistics, and Why Do We Need It?

Statistics is the science of collecting, organizing, presenting, analyzing and interpreting data. The word data sits at the heart of it all, so let's be clear about what data means.

Data are facts or information collected for a particular purpose. The number of pupils in your class, the scores you got in your last test, or the amount of rain that fell each day last week are all examples of data. Each is a set of facts gathered for a reason — to answer a question or find something out. Wherever facts are gathered for a purpose, statistics is at work.

Why do we need statistics? Because it helps us to do five important things:

(a) Collect information — bring facts together in one place.
(b) Organize information — put the facts in order so they are easy to read and handle.
(c) Compare information — look at two or more sets of facts side by side.
(d) Make decisions — use the facts to choose what to do.
(e) Present information clearly — show the facts in a way that others can understand.

These five are the purpose of statistics: it turns scattered facts into useful, clear information that helps us decide and act.

Example. Your class wants to know how many pupils walk to school.

First you collect information by asking every pupil how they travel. Then you organize information by counting how many walk, how many ride a bicycle, and how many are dropped off by car. Now you can compare information to see which way most pupils travel, and present it clearly in a small table. Finally, you can make a decision — for instance, whether the school should ask for more bicycle racks.

Every step you just took — collecting, organizing, comparing, presenting and deciding — is statistics.

Practice

1. What is statistics?
2. State what data are.
3. Your father records the amount he spends on food, transport and school fees each month, so he can see which costs him the most. Which two purposes of statistics is he using?
4. List the five ways that statistics helps us.

Section 2: Collecting, Tallying and Tabulating Data

When you want to find out something about a group, you first need to gather information about them. That gathered information is called data. There are four main ways to collect data:

1. Observation — you watch and note what you see, such as counting how many cars pass a spot.
2. Counting — you count items one by one.
3. Interview — you ask people questions and record their answers.
4. Questionnaire — you give people a written set of questions and collect their written answers.

Once you have collected data, you need a neat way to record it as it comes in. Tally marks are quick marks used to record data while you are collecting it. For every item you count, you draw one stroke. When you reach the fifth stroke, you draw it across the previous four to make a group like this: 𝍸 (four strokes with the fifth crossing them). Each group of five tally marks is easy to count later.

After tallying, you can set the data out in a frequency table. A frequency table is a table showing how many times each item occurs in a set of data. The column that lists each item's count is called the frequency.

Worked example — recording data with tally marks

Imagine you record the vehicles that pass by as: bus, car, car, car, lorry, bicycle, bicycle, car, car, lorry, bicycle, car.

To record this with tally marks, go through the list one item at a time. Start with the car column, adding a stroke for each car you meet. As you count the cars — car, car, car, car, car, car — you draw ||||/| (five) then one more, giving six tally marks for car. Continue for the other vehicles.

Your tally table looks like this:

Vehicles	Tally	Total
car		
bus		
lorry		
bicycle		
motor bike		

Read the last column (the Total) as the count for each item: car occurs 6 times, bus 1 time, lorry 2 times and bicycle 3 times. Notice the motor bike appears in the table with a total of 0 because it did not occur in the data.

Worked example — a frequency table

A class is asked their favourite colour, and the counts are collected. The results are set out in a frequency table:

Colour	Frequency
Red	18
Blue	20
Green	14
Yellow	16
White	15
Pink	30

The frequency column tells you how many students chose each colour. For instance, 30 students chose Pink and 14 chose Green. To check your table, you can add all the frequencies together to find the total number of students who answered.

So the whole process has three steps. First, collect the data using observation, counting, an interview or a questionnaire. Second, record the data with tally marks as it is collected. Third, tabulate — set the data out in a frequency table so you can read how many times each item occurs at a glance. Now try building a frequency table yourself.

Practice

1. State three methods of collecting data.
2. Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

Check your work

Attempt every question before you read this page.

What is Statistics, and Why Do We Need It?

1. Statistics is the science of collecting, organizing, presenting, analyzing and interpreting data.
2. Data are facts or information collected for a particular purpose.

3. He is collecting information (recording the amounts) and comparing information (seeing which cost is the most).

   Recording the amounts each month is collecting information. Looking at the amounts to see which is the largest is comparing information. He could also be making a decision about where to spend less.
4. It helps us to (a) collect information, (b) organize information, (c) compare information, (d) make decisions, and (e) present information clearly.

Collecting, Tallying and Tabulating Data

1. Any three of: observation, counting, interview, questionnaire.
2.

Number	Frequency
1	3
2	3
3	4
4	3

   Go through the list and count how many times each number appears. The number 1 appears three times (1, 1, 1), 2 appears three times (2, 2, 2), 3 appears four times (3, 3, 3, 3) and 4 appears three times (4, 4, 4). Write each number once in the first column and its count in the frequency column.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-603a66b9	**Date**	
Subtopic	Need for statistics: purposes, and the collection and tabulation of data	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____